# From Linear Models to Modern Sequence Models

## A Comparative Study for Fashion-MNIST Image Classification

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# Problem and research questions

## Task

Classify one $28 \times 28$ grayscale apparel image into one of 10 Fashion-MNIST categories. Compare five implemented architectures under a shared split and evaluation protocol.

1. How do Linear, MLP, CNN, GRU and Transformer compare on held-out accuracy and macro-F1?
2. How do flattened, spatial, row-sequence and patch-sequence representations relate to class-level strengths and errors?
3. What quality–cost trade-offs appear across parameter count, T4 training time and measured CPU inference?
4. How consistent are the results across three run seeds?

We compare these specific configurations; representation, capacity and training policy vary together, so this is not a causal architecture ablation.

# Dataset: source, format and labels

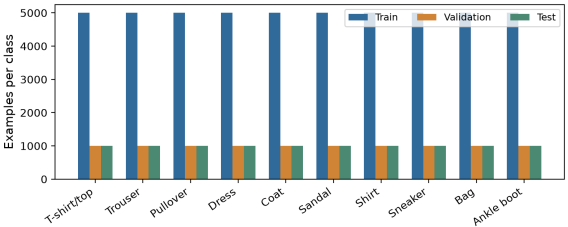
- Fashion-MNIST is a Zalando Research benchmark: 60,000 official training images and 10,000 official test images.
- Each image is a **28×28, 8-bit grayscale** catalog thumbnail with one of ten labels.
- Original dataset construction trims borders, centers and converts catalog images to grayscale; the project loads original IDX files through torchvision.
- The dataset is MIT-licensed; CSV exports are not used by the training pipeline.
- Fixed stratified split (seed 36): 50,000 train / 10,000 validation; official test remains separate. Each class contributes 5,000 / 1,000 / 1,000 examples.

## Ten apparel classes

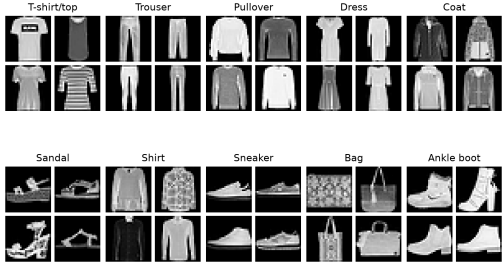
| | | | |
|---|---|---|---|
| 0 | T-shirt/top | 5 | Sandal |
| 1 | Trouser | 6 | Shirt |
| 2 | Pullover | 7 | Sneaker |
| 3 | Dress | 8 | Bag |
| 4 | Coat | 9 | Ankle boot |

Each split is balanced by class. Visual similarity among tops, shirts, pullovers and coats remains despite equal support.

# Exploratory data analysis: class balance and examples




Train / validation / official-test class counts.




Forty representative training images: four examples per class.

## EDA findings

Largest-to-smallest class-support ratio is **1.0** in each partition; no resampling or class weighting is needed. Repeated upper-body silhouettes suggest plausible confusions, but no manual relabeling or perceptual-duplicate audit was performed.

# Preprocessing, split controls and batch contract

## Training path

1. Read the persisted seed-36 stratified split.
2. Pad each image by 2 pixels with zeros; take a random $28 \times 28$ crop (training only).
3. Convert uint8 pixels to float32 in $[0, 1]$, then normalize: $x' = (x - \mu_{train})/\sigma_{train}$.

## Validation and test

Tensor conversion and the same train-derived normalization only; no random crop or other augmentation. Test metrics never select checkpoints.

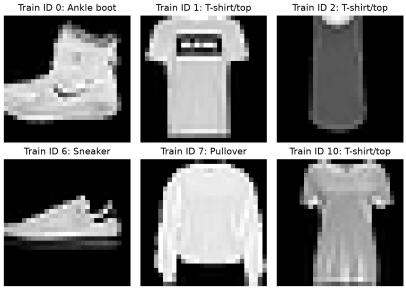
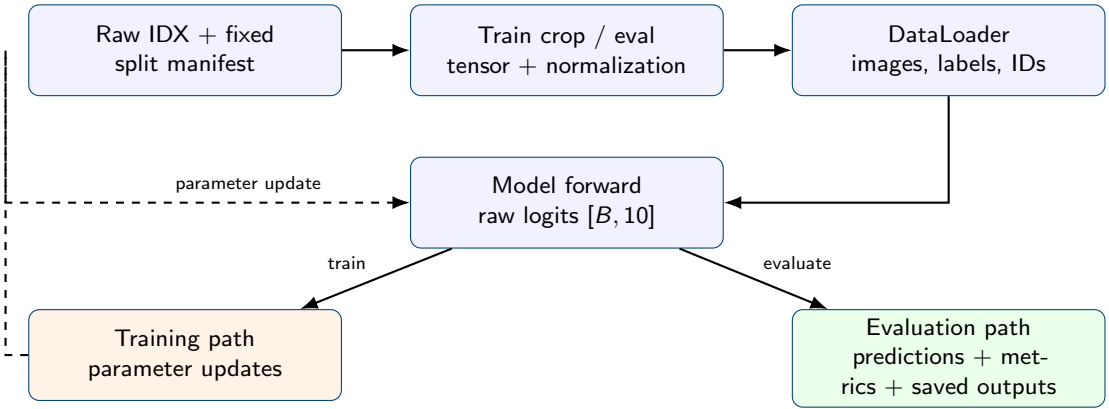


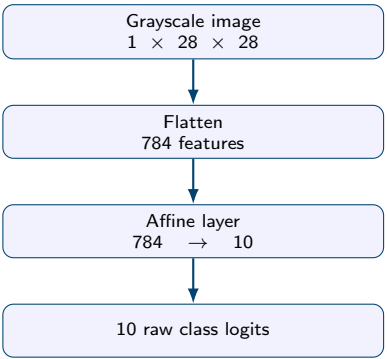

Illustration of a normalized training batch; augmentation is applied to training samples only. Every model receives float32 $[B, 1, 28, 28]$, labels $[B]$, and stable sample IDs. Batch size is 64; incomplete final batches are retained.

# End-to-end data pipeline




Both paths share the same image, label and sample-ID batch contract. Training loops back through parameter updates; evaluation produces outputs without changing model parameters.

# Model 1: Linear / softmax classifier




**7,850 parameters.** No hidden layer or nonlinearity; the baseline learns one affine score for each class.

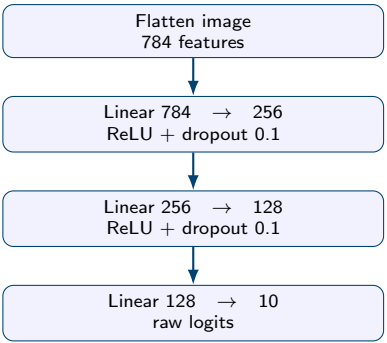
## Representation

Flatten each image to 784 values, then apply one affine map to ten logits. This model has no hidden layer or nonlinearity.

## Configuration

7,850 trainable parameters. The linear model is the pixel-wise baseline for the nonlinear alternatives.

# Model 2: Multilayer perceptron (MLP)




**235,146 parameters.** Dense nonlinear features combine pixels globally; dropout regularizes hidden activations during training.

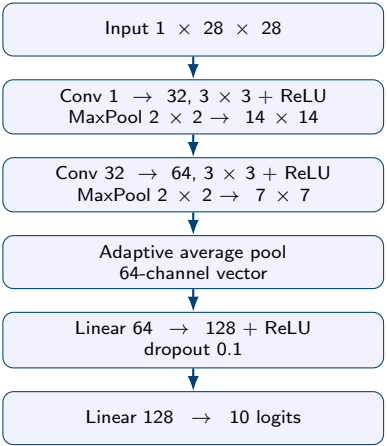
## Representation

Two dense hidden layers learn global pixel combinations. ReLU enables nonlinear boundaries; dropout 0.1 regularizes hidden activations during training.

## Configuration

235,146 trainable parameters; dropout is disabled during evaluation. The final layer returns ten raw logits.

# Model 3: Custom convolutional neural network (CNN)




**28,426 parameters.** Shared local filters learn spatial feature maps; pooling reduces resolution.

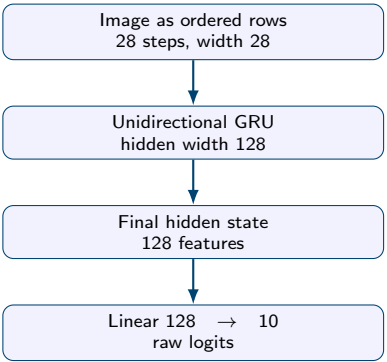
## Representation

Learned $3 \times 3$ filters share weights over local neighborhoods and create feature maps. Two max-pooling stages reduce spatial size $28 \to 14 \to 7$; adaptive pooling yields 64 features.

## Configuration

28,426 trainable parameters. The custom-from-scratch head is $64 \to 128 \to 10$, with ReLU and dropout 0.1.

# Model 4: Recurrent model (GRU)




**61,962 parameters.** Each timestep is one full image row; recurrent state carries context across rows.

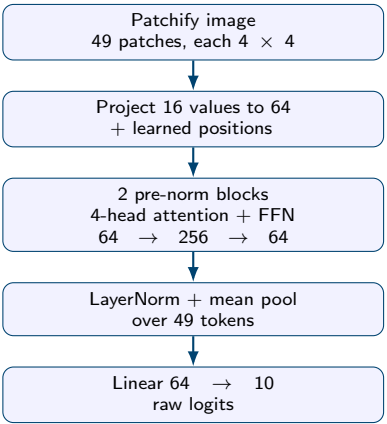
## Representation

Each image row is one timestep of width 28. A single unidirectional GRU carries hidden state across 28 ordered rows; its final state feeds the classifier.

## Configuration

61,962 trainable parameters. Column traversal, bidirectionality and LSTM variants were not evaluated.

# Model 5: Patch Transformer




**104,970 parameters.** Every patch token can attend to all 49 positioned tokens; no class token is used.

## Representation

A $7 \times 7$ patch grid becomes 49 position-aware tokens. Two pre-norm blocks use four attention heads and an FFN $64 \to 256 \to 64$ to mix token context.

## Configuration

104,970 trainable parameters. LayerNorm and mean pooling feed a $64 \to 10$ head; no class token or pretrained encoder is used.

# Methodology: training, evaluation and checkpoints

## Loss and metrics

Every model minimizes unweighted multiclass cross-entropy on raw logits:

$$L = -\frac{1}{N}\sum_{i} \log \frac{e^{z_{i,y_i}}}{\sum_{k=0}^{9} e^{z_{i,k}}}.$$

No softmax is applied before the loss. Evaluation records probabilities and predicts by argmax. Report accuracy, macro-F1 and cross-entropy; macro-F1 weights each class equally.

## Optimization and selection

- Adam, learning rate $10^{-3}$, weight decay 0, batch 64, up to 100 epochs.
- Linear/MLP: cosine schedule to $10^{-5}$, early stop (patience 15; F1 min delta 0.001).
- CNN/GRU/Transformer: no scheduler; no early stopping.
- Select `best.pt` by validation macro-F1; ties: lower validation loss, then earlier epoch. Test never used for selection.

All 15 runs preserve configs, source/environment metadata, histories and best/last checkpoints; last checkpoints support epoch-boundary resumption.

# Experimental setup and reproducibility

## Training and evaluation

- Five configurations $\times$ three seeds: **69420, 67, 69** (15 runs).
- Training: Colab Linux, NVIDIA Tesla T4; PyTorch 2.11.0+cu130, torchvision 0.26.0+cu130, CUDA 13.0.
- Float32, deterministic algorithms enabled; 100-epoch cap.
- Official test: 10,000 images per checkpoint; report mean $\pm$ sample SD across seeds.

## Timing protocols

- Training time: summed train-and-validation epoch loops on T4; excludes startup/checkpoint overhead.
- Inference: separate Windows CPU; Python 3.14.6, PyTorch 2.13.0+cpu, torchvision 0.28.0+cpu; 2 threads.
- Batch 64, float32, 20 warm-ups + 100 forward batches; excludes loading/transfer and is not batch-one latency.

One effective configuration per family; no exhaustive search or matched-compute/parameter ablation. Full run/checkpoint identities and reproduction instructions are retained with project artifacts.

# Results: held-out performance

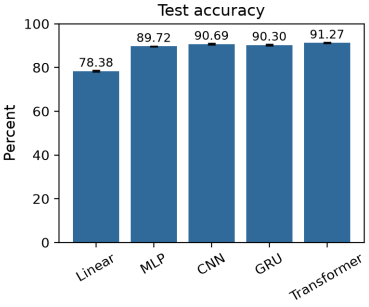
Mean ± sample SD across three seeds; validation guides checkpoint selection, official test is final evaluation.

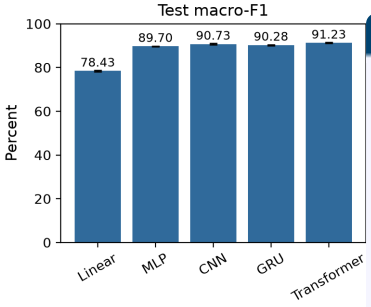
| | Validation | | | Official test | | |
|---|---|---|---|---|---|---|
| Model | Acc. (%) | Macro-F1 (%) | CE | Acc. (%) | Macro-F1 (%) | CE |
| Linear | 79.267±0.176 | 79.254±0.159 | 0.5867 | 78.377±0.283 | 78.427±0.307 | 0.6114 |
| MLP | 90.333±0.165 | 90.288±0.157 | 0.2649 | 89.720±0.104 | 89.698±0.094 | 0.2881 |
| CNN | 91.547±0.115 | 91.556±0.056 | 0.2415 | 90.690±0.324 | 90.726±0.262 | 0.2642 |
| GRU | 91.063±0.061 | 91.055±0.064 | 0.2655 | 90.297±0.255 | 90.281±0.250 | 0.2836 |
| Transformer | **91.580±0.026** | **91.548±0.011** | **0.2370** | **91.267±0.237** | **91.234±0.215** | **0.2444** |

## Main result

Transformer leads mean test accuracy and macro-F1; CNN is close behind at 90.690% / 90.726%. The gap is **0.577 / 0.508 percentage points**, not proof of universal superiority.

# Results: quality comparison




## What changes?

- MLP gains **11.343 points** test accuracy over Linear.
- CNN and Transformer exceed 90% test accuracy and macro-F1.
- Transformer beats CNN on test accuracy in all 3 paired seeds: +0.81, +0.62, +0.30 points.
- Across models, test-accuracy SD is 0.104–0.324 points.

Only three seeds; paired differences do not establish statistical significance.

# Results: parameters and compute trade-offs

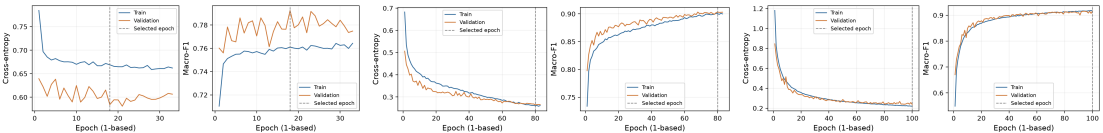
| Model | Parameters | T4 loop (min) | CPU (ms/image) |
|---|---|---|---|
| Linear | 7,850 | 10.27±1.05 | 0.00152±0.00005 |
| MLP | 235,146 | 29.85±3.18 | 0.01799±0.00202 |
| CNN | 28,426 | 33.49±0.44 | 0.40430±0.04161 |
| GRU | 61,962 | 32.89±0.24 | 0.10777±0.03218 |
| Transformer | 104,970 | 38.39±0.91 | 0.44474±0.05600 |

## Quality–cost takeaway

CNN stays close to Transformer quality with **72.9% fewer parameters** and a **4.90-minute lower** observed T4 loop total. GRU's measured CPU latency is intermediate; MLP has many more parameters but does not lead in quality.

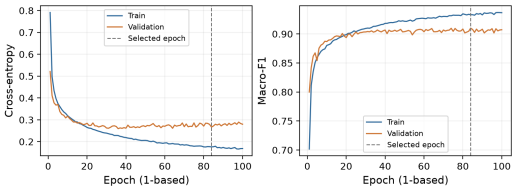
Training means train + validation epoch-loop time. CPU is a separate Windows batch-64 forward-only profile; timings are machine-specific, not end-to-end latency.

# Results: learning dynamics

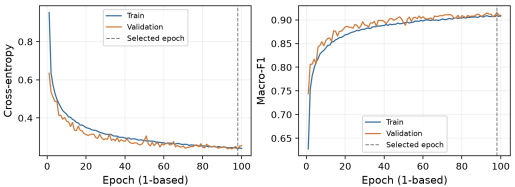



Seed 67; vertical marker is the selected validation checkpoint. Linear plateaus well below nonlinear models.

# Results: learning dynamics




GRU




Transformer

Seed 67; GRU shows late train/validation divergence. Training metrics include augmentation and training-mode behavior.

# Training-curve analysis

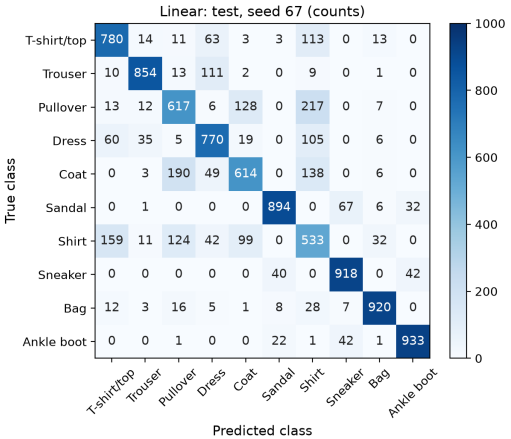
- **Linear**: validation F1 improves early, then fluctuates and stabilizes below nonlinear models; selected seed-67 checkpoint is epoch 18.
- **MLP**: rapid early gain followed by smaller late improvements; selected epoch 80.
- **CNN**: high validation F1 with comparatively close train/validation curves; selected epoch 100 for seed 67.
- **GRU**: clearest late divergence—training F1 continues upward while validation F1 plateaus and validation loss rises slightly; selected epoch 84 of 100.
- **Transformer**: rapid early improvement followed by smaller late gains; selected epoch 98.

## Interpretation

Training metrics include augmentation and training-mode behavior, while validation uses deterministic preprocessing and evaluation mode. The GRU pattern is consistent with diminishing validation gains but does not identify a causal mechanism.

Learning histories shown on the preceding slides are for seed 67; checkpoint selection uses validation macro-F1.

# Confusion matrix: Linear

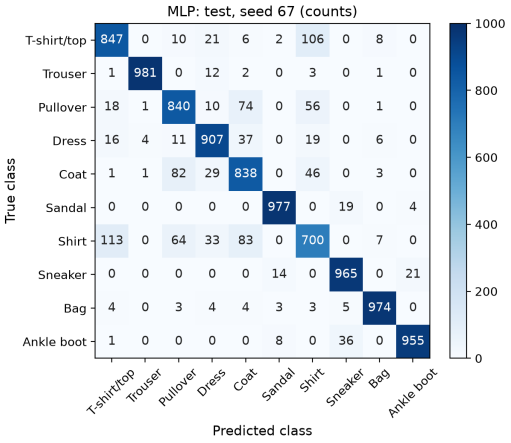



Official-test confusion counts; 1,000 images per class.

## Seed-67 class errors

- Shirt is weakest (F1 0.497).
- 159 Shirts map to T-shirt/top, 124 to Pullover and 99 to Coat.
- Pullover/Coat confusions are also frequent: 128 and 190 in opposite directions.

# Confusion matrix: MLP

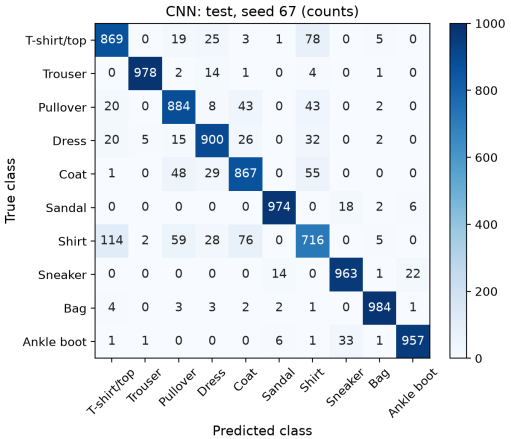



Official-test confusion counts; 1,000 images per class.

## Seed-67 class errors

- Trouser F1 is 0.987; Shirt remains lowest at 0.724.
- Shirt → T-shirt/top: 113 errors; reverse direction: 106.
- Pullover/Coat confusion persists (74 and 82).

# Confusion matrix: CNN

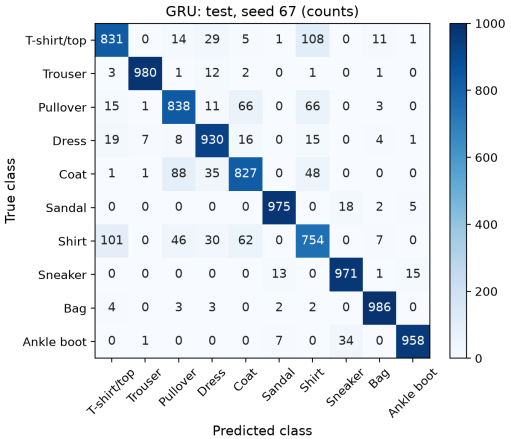



## Seed-67 class errors

- Overall accuracy is 90.92%; Shirt is weakest (F1 0.742).
- Shirt errors: 114 to T-shirt/top, 76 to Coat and 59 to Pullover.
- Pullover/Coat errors (43 and 48) are lower than MLP's in this run.

Official-test confusion counts; the comparison does not isolate a causal effect of local convolutions.

# Confusion matrix: GRU

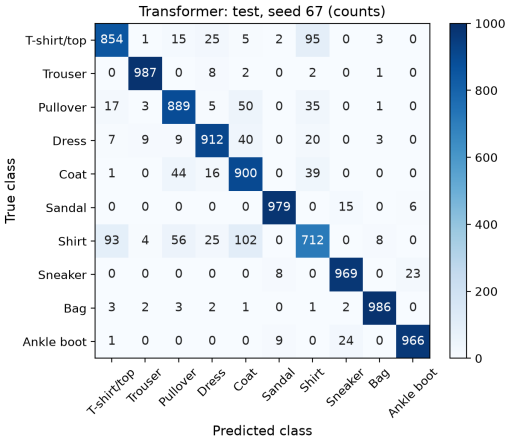



Official-test confusion counts; 1,000 images per class.

## Seed-67 class errors

- Overall accuracy is 90.50%; Bag recall reaches 0.986 and Dress recall 0.930.
- Shirt recall is highest among the five models (0.754), though Shirt remains its weakest-F1 class (0.756).
- Aggregate ranking does not capture every class-level strength.

# Confusion matrix: Transformer

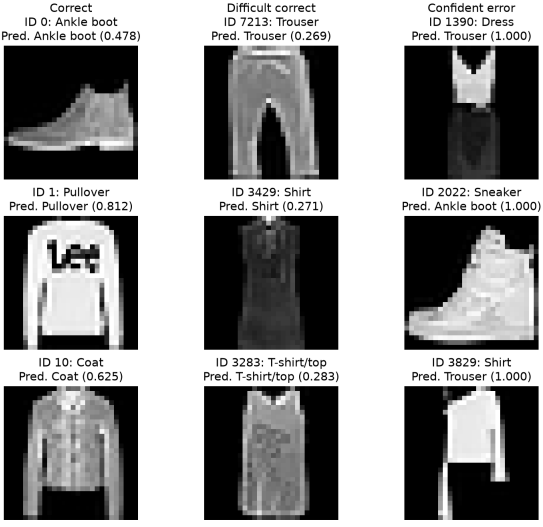



## Seed-67 class errors

- Overall accuracy is 91.54%; Bag and Trouser F1 are 0.985 and 0.984.
- Shirt remains weakest (F1 0.748); recall is 0.712.
- Shirt errors: 102 to Coat, 93 to T-shirt/top and 56 to Pullover.

Official-test confusion counts; aggregate performance summarizes all three seeds.

# Prediction gallery: Linear



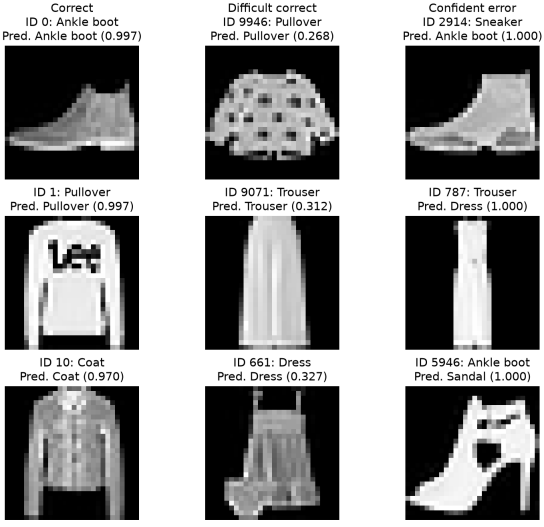

Selected seed-67 test examples: common correct, low-confidence correct and high-confidence incorrect; not a prevalence estimate.

# Prediction gallery: MLP



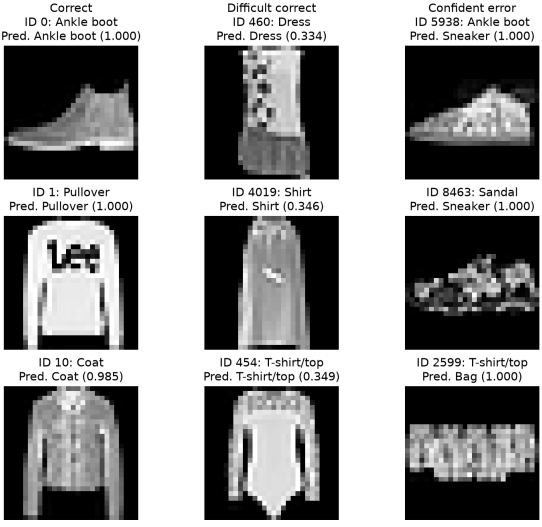

Selected seed-67 test examples: common correct, low-confidence correct and high-confidence incorrect; not a prevalence estimate.

# Prediction gallery: CNN



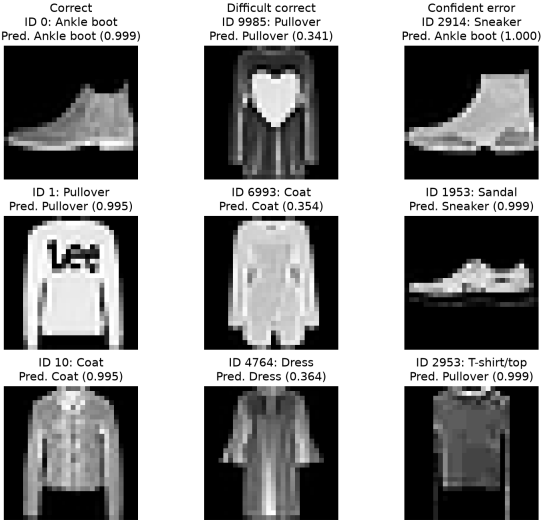

Selected seed-67 test examples: common correct, low-confidence correct and high-confidence incorrect; not a prevalence estimate.

# Prediction gallery: GRU




Selected seed-67 test examples: common correct, low-confidence correct and high-confidence incorrect; not a prevalence estimate.

# Prediction gallery: Transformer




Selected seed-67 test examples: common correct, low-confidence correct and high-confidence incorrect; not a prevalence estimate.

# Discussion

## Interpreting the comparison

- Adding nonlinear capacity yields a large improvement over the flattened linear baseline; the MLP gains 11.343 test-accuracy points.
- The patch Transformer has the strongest mean test scores, while the CNN is close behind with substantially fewer parameters.
- Architecture rankings do not describe every class equally: the GRU has the strongest seed-67 Shirt recall, despite not leading aggregate metrics.
- Shirt and visually similar upper-body garments remain difficult across all five models, indicating a shared ambiguity in the benchmark.

These observations describe the evaluated configurations; representation, parameter count and training policy vary together.

# Limitations

## Scope of the evidence

- Results use one fixed train/validation split and three run seeds; no formal significance test was conducted.
- Each model family has one configuration. Models are not matched by parameter count or compute, and no controlled ablation isolates architecture effects.
- Evaluation is limited to Fashion-MNIST's small grayscale images; there is no external-dataset, robustness or perceptual-duplicate evaluation.
- Training and CPU inference timings come from different machines and software environments, so they are machine-specific measurements rather than deployment latency.
- Seed-67 confusion matrices and prediction galleries illustrate individual-run behavior; aggregate performance is reported separately across all three seeds.

# Conclusion

## Main finding

On this Fashion-MNIST setup, the Transformer achieves the highest mean test accuracy (91.267%) and macro-F1 (91.234%). The CNN is a strong compact alternative at 90.690% accuracy and 90.726% macro-F1, using 72.9% fewer parameters.

## Takeaway

Nonlinear image models outperform the linear baseline, but the observed Transformer lead is configuration-specific—not evidence of universal architectural superiority. Similar apparel silhouettes, especially Shirt and related tops, remain a common source of errors.

# Thank you

Questions?

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